

Reading a Polynomial: Terms, Factors, Coefficients, Degree

Answer Key

Algebra 1

Quick Answers

- | | |
|---------------------------|------------------------|
| 1. -11 | 6. -8 |
| 2. $x^2 + 8x + 12$ | 7. $<$ |
| 3. $6x^2(3x + 5)$ | 8. $(x - 7)(x + 4)$ |
| 4. $(a) = 6x^2 - 15x$, — | 9. $(a) = 12x$, — |
| 5. $2x^2 - 7x - 15$ | 10. $(a) = 3x + 2$, — |
-

What is verified

7 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 9, 10. The worked solution states what a correct response must contain.

The author marked 12 blanks as working space (*not* answers — nothing checks what lands there).

Curriculum

Level: Algebra 1

HSA-APR.A.1 — *problems 1, 2, 4, 5, 6, 7, 9*

HSA-SSE.B.3 — *problems 3, 8, 10*

Difficulty 1–5 of 5 across 13 tagged checks.

1. $P(x) = 4x^3 - 7x^2 + x - 9$: complete the term/coefficient/degree table, then find $P(1)$.

Solution: Reading each term off in turn:

Term	Coefficient	Degree of the term
$4x^3$	4	3
$-7x^2$	-7	2
x	1	1
-9	-9	0

Two cells are the ones students miss: the x term has coefficient 1 (not “none”), and the constant -9 has degree 0, because $-9 = -9x^0$. The degree of P is the largest of those term degrees, 3. Substituting $x = 1$ turns every power of x into 1, so $P(1)$ is just the sum of the four coefficients: $4 + (-7) + 1 + (-9)$.

$P(1) = -11$

- 2.** Area model for $(x + 6)(x + 2)$: fill in the four partial areas and write the total area in standard form.

Solution: Each cell is (row label) $\times$ (column label):

$\times$	x	$+6$
x	x^2	$6x$
$+2$	$2x$	12

The two shaded-diagonal cells $6x$ and $2x$ are like terms, so they combine: $x^2 + 6x + 2x + 12 = x^2 + 8x + 12$. The four cells of the model are exactly the four partial products, and the middle coefficient 8 is the sum of the two numbers in the binomials.

$$x^2 + 8x + 12$$

- 3.** Write $18x^3 + 30x^2$ as (greatest common monomial factor) $\times$ (remaining factor).

Solution:

$$18 = 2 \cdot 3 \cdot 3, \quad 30 = 2 \cdot 3 \cdot 5 \quad \Rightarrow \quad \gcd(18, 30) = 6,$$

and the lowest power of x present is x^2 , so the common monomial factor is $6x^2$:

$$18x^3 + 30x^2 = 6x^2 \cdot 3x + 6x^2 \cdot 5 = 6x^2(3x + 5).$$

Check by re-distributing: $6x^2 \cdot 3x = 18x^3$ and $6x^2 \cdot 5 = 30x^2$.

$$6x^2(3x + 5)$$

- 4.** $3x(2x - 5)$ versus $6x^2 - 15x$: (a) confirm they are equivalent; (b) which form shows what?

Solution (a): Distribute $3x$ across the binomial: $3x \cdot 2x = 6x^2$ and $3x \cdot (-5) = -15x$, so $3x(2x - 5) = 6x^2 - 15x$.

$$6x^2 - 15x$$

Solution (b) — instructor-judged. The product form $3x(2x - 5)$ shows the *factors*: it says at a glance that the polynomial is divisible by $3x$ and by $(2x - 5)$, which is what you need when hunting for zeros. The standard form $6x^2 - 15x$ shows the *degree* (2, from the leading term) and the *coefficients* (6 and -15). Full credit: names one thing each form makes visible. Half credit: describes only one form.

- 5.** Multiply $(2x + 3)(x - 5)$ and write it in standard form.

Solution:

$$\begin{aligned} (2x + 3)(x - 5) &= 2x \cdot x + 2x \cdot (-5) + 3 \cdot x + 3 \cdot (-5) \\ &= 2x^2 - 10x + 3x - 15 \\ &= 2x^2 - 7x - 15. \end{aligned}$$

Read off the parts: leading coefficient 2, coefficient of x equal to -7 (from $-10 + 3$), constant term -15 , degree 2.

$$2x^2 - 7x - 15$$

- 6.** $Q(x) = (x + 4)(x - 1)(x + 2)$: find the constant term using $Q(0)$.

Solution: Every term of the expanded product except the constant one carries at least one factor of x , so substituting $x = 0$ wipes them all out and leaves the constant term alone. Substituting into the factored form is far less work than expanding: $Q(0) = (0 + 4)(0 - 1)(0 + 2) = (4)(-1)(2)$. $Q(0) = -8$

The constant term of Q is therefore -8 , and Q has degree 3 because three linear factors are multiplied together.

7. $A(x) = (3x^2 - 1)(x + 5)$, $B(x) = 2x^4 - x^2$: compare the degrees.

Solution: The degree of a product is the sum of the degrees of its factors, so A has degree $2 + 1 = 3$; there is no need to expand it. For B , the degree is the largest exponent that appears, which is 4. Since 3 is less than 4: $<$

8. Factor $x^2 - 3x - 28$ into two binomials.

Solution: Look for two numbers whose product is the constant term -28 and whose sum is the coefficient of x , namely -3 . The pair -7 and $+4$ works: $(-7)(4) = -28$ and $(-7) + 4 = -3$. Therefore $x^2 - 3x - 28 = (x - 7)(x + 4)$. $(x - 7)(x + 4)$

9. $(x + 3)^2 - (x - 3)^2$: (a) expand and simplify; (b) explain the drop in degree.

Solution (a):

$$\begin{aligned}(x + 3)^2 - (x - 3)^2 &= (x^2 + 6x + 9) - (x^2 - 6x + 9) \\ &= x^2 + 6x + 9 - x^2 + 6x - 9 \\ &= 12x.\end{aligned}$$

$12x$

Solution (b) — instructor-judged. Each square contributes an x^2 term with coefficient $+1$, so the subtraction cancels them exactly; the constants $+9$ cancel too. What survives is the difference of the linear terms, $6x - (-6x) = 12x$, of degree 1. Full credit must point at the cancelling x^2 terms, not merely restate the answer.

10. Area $6x^2 + 19x + 10$ with one side $(2x + 5)$: (a) find the other side; (b) why do the degrees add?

Solution (a): The other side is the factor that pairs with $(2x + 5)$. Split the middle term using two numbers with product $6 \cdot 10 = 60$ and sum 19, namely 4 and 15:

$$\begin{aligned}6x^2 + 19x + 10 &= 6x^2 + 4x + 15x + 10 \\ &= 2x(3x + 2) + 5(3x + 2) \\ &= (2x + 5)(3x + 2).\end{aligned}$$

Dividing the area by the known side leaves the other side.

$3x + 2$

Solution (b) — instructor-judged. Multiplying two polynomials multiplies their leading terms, so the degrees add: $1 + 1 = 2$. A degree-2 area built from a degree-1 side therefore has to leave a degree-1 side. Full credit states that add-the-degrees rule rather than only reciting the numbers.